

Policy Conflict Internals in Real Agentic Contexts

Working Typst draft. Prose target: under 2000 words, excluding tables, captions, and TODO blocks.

TODO

Write the final opener. Keep the claim narrow: this is not a production detector or causal mechanism claim. The strongest story is that synthetic policy-conflict directions can recover a shape-specific real production signal after the ontology is narrowed.

Problem

This work follows up on our recent DX Terminal collaboration, where we used mechanistic interpretability in real-world financial contexts. In part 1, we looked for early signs of interpretable market-perception geometry in LLM activations over real market data.

Part 2 looks at a different problem DXRG saw in their agents: strange behavior when policy sources collide. A trading agent sees system rules, user strategies, active vault settings, portfolio state, market data, and prior decisions. Those sources are not always mutually consistent.

TODO

Add one concrete DX Terminal example in plain language: strategy says buy aggressively while settings cap size; strategy says sell all while settings imply partial exits; complaint refers to a strategy no longer active.

Our question was whether these policy-source conflicts leave a readable internal signal. The useful version of that question is not “can we classify all complaints?” It is narrower: can we train clean synthetic directions for specific conflict shapes, then see whether those directions fire on analogous real prompts?

Why This Matters

One of Concordance’s goals is to make interpretability useful to agent builders working on real systems. The DX Terminal policy-conflict problem was valuable because it was concrete, recurring, and operationally meaningful: if an agent’s internal state reflects competing constraints, that signal could eventually support monitoring, audits, prompt redesign, settings semantics, and strategy lifecycle debugging.

TODO

Add your voice here. The current paragraph is deliberately functional. You may want one paragraph on why “real data -> synthetic abstraction -> real data” is the loop this post is trying to demonstrate.

Synthetic Abstraction

Real prompts are entangled, so we started with controlled prompt families. The synthetic prompts isolate conflicts between user-configured strategies and vault-configured active settings while keeping the decision surface simple.

SOURCE

Dataset and workflow sources: phase_09/scripts/build_phase_09_dataset.py for trade_size, phase_10/scripts/build_phase_10_dataset.py for risk_preference, phase_12/scripts/build_phase_12_dataset.py for diversification_preference, and phase_12/specs/workflow.py plus phase_12/specs/marshalls_strict_workflow.py for lexical holdout probes.

```
[system]
Role: trading agent.
Core rule: each prompt contains STRATEGY and ACTIVE SETTINGS.
Priority rule: ACTIVE SETTINGS are binding execution constraints.
Decision order: activity, asset/risk/diversification posture, then size.

[user]
TASK: choose exactly one action for this tick.
STRATEGY: compact user preference.
ACTIVE SETTINGS: slider-like constraints.
PORTFOLIO: controlled portfolio state.
MARKET: controlled synthetic assets and evidence.
Output: strict JSON.
```

We tested three families:

- `trade_size`: buy small vs large, with the output size/action as the axis.
- `risk_preference`: asset selection by allowed risk posture.
- `diversification_preference`: concentration vs broadening, conditioned on the portfolio.

Each family used paraphrased strategy templates and settings-label templates. Those templates were split into train/test groups, then probes were evaluated on held-out strategy wording, held-out settings wording, and stricter both-axis holdouts. The point was to test the conflict relation rather than a phrase family.

TODO

Optional: replace the prompt sketch with a more narrative paragraph. The current version is data-dense and edit-friendly, but not yet blog-polished.

Synthetic Probe Results

After iterating on prompt structure and confound controls, all three families were linearly readable under lexical holdouts. Values below are balanced accuracy / AUROC where paired.

SOURCE

Canonical numbers: `phase_12/reports/PROMPT_CONFLICT_FAMILY_CHECKPOINT_2026_04_16.typ`. Blog-friendly figures and captions: `phase_12/reports/DX_TERMINAL_CONFLICT_GEOMETRY_BRIEF_2026_04_17.typ`.

Family	Standard probe results	Strict holdout
<code>trade_size</code>	XOR 0.9948 / 1.0000; strategy holdout 1.0000 / 1.0000; settings holdout 0.9948 / 1.0000	0.990 / 1.000 at L40
<code>risk_preference</code>	XOR 0.9635 / 0.9766; strategy holdout 0.9844 / 0.9937; settings holdout 0.9740 / 0.9839	0.8854 / 0.9119
<code>diversification_preference</code>	behavior: aligned 1.0000, conflict 0.8542; XOR 0.9896 / 0.9995; strategy holdout 1.0000 / 1.0000; settings holdout 0.9792 / 0.9957	0.8333 / 0.8819

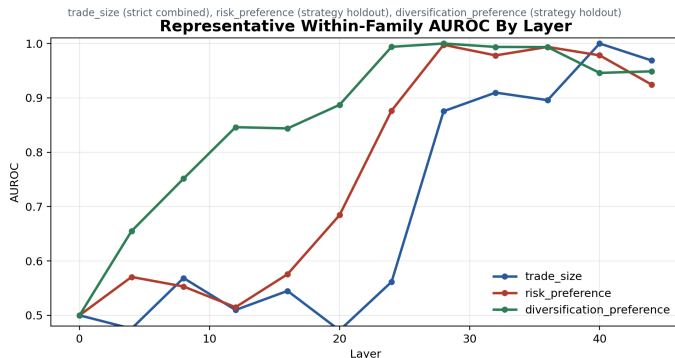


Figure 1: Representative within-family AUROC curves across layers.

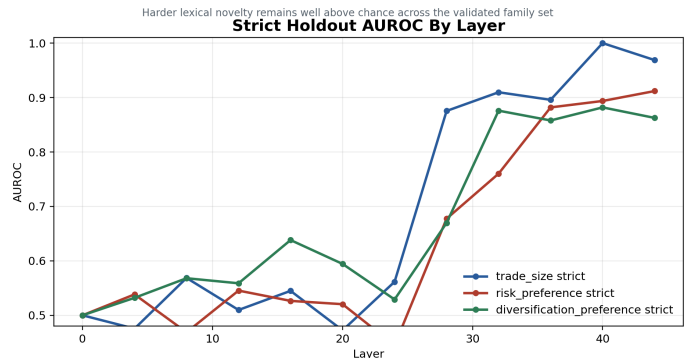


Figure 2: Strict lexical-holdout AUROC curves.

The families are related but not identical. Same-capture mean-difference directions at L36 showed moderate cosine similarity, strongest between `risk_preference` and `trade_size`.

Pair	L36 cosine
risk_preference vs trade_size	0.6449
diversification_preference vs risk_preference	0.4684
diversification_preference vs trade_size	0.4883

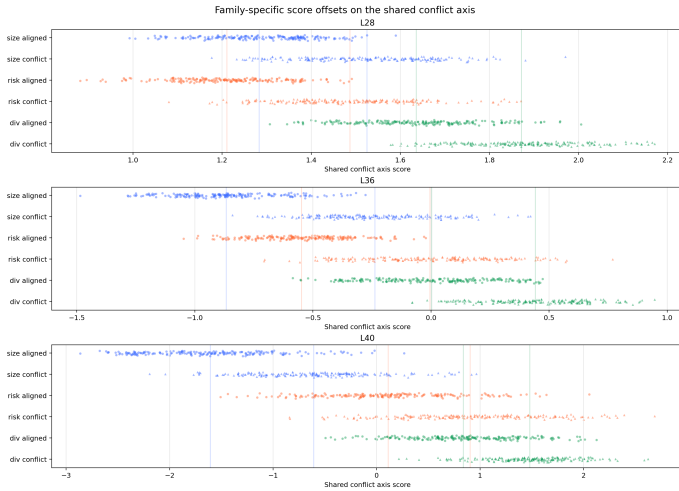


Figure 3: Shared-axis distributions: separation exists, but family baselines are offset.

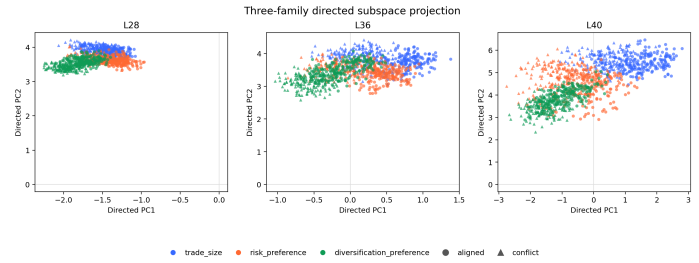


Figure 4: Directed subspace view: related conflict geometry, not one collinear axis.

TODO

Write the transition from “synthetic readout works” to “transfer is hard.” Important claim boundary: high AUROC supports a representational readout, not a causal mechanism.

First Real Transfer Attempt

The first direct transfer pass projected synthetic conflict directions onto full production prompts at coarse global sites. It did not cleanly separate complaint rows from baseline controls.

SOURCE

Bridge source: phase_12/reports/REAL_TRANSFER_BRIDGE_PLAN_2026_04_22.md. The plan records the initial null/unclear full-prompt transfer and frames the next experiments as tests of concept failure, format failure, and site failure.

That failure mattered. It showed that synthetic success does not automatically become production success. Real prompts differ in template, label semantics, time horizon, complaint noise, and where the relevant evidence appears inside the prompt.

TODO

Add one paragraph with your interpretation of the failure. A good frame: this was not “the probe failed,” it was the point where the ontology had to get sharper.

Bridge Program

We then used bridge datasets to separate template mismatch from content and ontology mismatch.

Dataset	Rows	Aligned	Conflict
Stage 1a template control	768	384	384
Stage 1b loose adapter	258	168	90
Stage 1b strict adapter	118	81	37

Stage 1b strict buy-only	33	27	6
--------------------------	----	----	---

SOURCE

Neon tables: dx_terminal_trade_size_stagela_template_control_v1, dx_terminal_trade_size_stagelb_adapter_loose_v1, dx_terminal_trade_size_stagelb_adapter_strict_v1, dx_terminal_trade_size_stagelb_adapter_strict_buy_only_v1. Summary JSONs live in phase_12/outputs/transfer_bridge/.

The bridge evidence was real but weak. Stage 1b was noisy; buy-only filtering helped, but probe-to-synthetic cosine remained near zero. Sell/liquidation contamination was part of the problem, but the deeper issue was an unresolved ontology and representation mismatch.

TODO

Decide how much bridge detail belongs in the final post. If the post is for a broad audience, this section can be a short failure-analysis interlude before Phase 13.

Phase 13 Real Signal Discovery

Phase 13 asked a simpler question: if we do not train a classifier and do not set thresholds, do fixed synthetic directions produce scalar structure anywhere on real DX Terminal prompts?

Medium validation run: wr_14f78308dbac_dbc78513. Corpus: dx_terminal_signal_discovery_phase13_v1. Scope: aggressive tier, end-of-section sites, 500 complaints, 300 structure-matched controls, 118 Stage 1b strict anchors, and 33 buy-only anchors. Primary cell: L32 settings_end.

SOURCE

Phase 13 source: phase_13/reports/PHASE13_REAL_TRANSFER_SIGNAL_BRIEF_2026_04_24.typ.

Direction	Anchor	Complaint	Structure control	Anchor-control	Complaint-control
trade_size	4.425	3.803	3.278	+1.147	+0.526
shared_mean	3.462	3.137	2.760	+0.703	+0.377

At this cell, trade_size and shared_mean separated complaint prompts from structure-matched controls. risk_preference was weaker and diversification_preference was not clean, so this is not a generic “all synthetic directions light up on all complaints” result.

TODO

Add the intuitive explanation of L32 settings_end: by this point the model has seen ACTIVE SETTINGS, but has not yet consumed the whole prompt.

Row Reading And Ontology Correction

The most important interpretability move was reading the top and bottom rows. The preregistered proxy expected USER_CONFIG_CONFLICT to rank high and RULE_FABRICATION or non-config rows to rank low. That proxy was wrong for the trade_size target. Root-cause labels diagnose why a complaint happened; the probe target is visible current-prefix conflict shape.

Direction	Top action/size	Top strategy ignored	Bottom action/size	Bottom strategy ignored
trade_size	20/25	5/25	15/25	10/25
shared_mean	20/25	5/25	9/25	16/25

Top trade_size complaint type	Count
UNWANTED_BUY	10/25
UNWANTED_SELL	6/25

WRONG_SIZE	4/25
Concrete action/size combined	20/25

High `trade_size` rows were mostly concrete current action or size complaints: “why did you buy HOTDOGZ?”, “why did you buy so much POOPCOIN?”, or “Buy available balance 30%, not 10 ETH.” Low rows could still be valid complaints, but often depended on history, lifecycle state, prior strategy fulfillment, or rule interpretation rather than a clean current-prefix size conflict.

TODO

Expand this into the core narrative section. The lesson is strong: looking at internals forced the label ontology to change from root cause to conflict shape.

Claim Boundary

CLAIM

Fixed synthetic directions recover a real production signal at L32 `settings_end`. `trade_size` is selective for current-prefix concrete sized-action conflict. `shared_mean` appears to track broader policy tension, but the shared-family interpretation still needs more audit.

This is not a final detector, not a deployment monitor, and not a causal mechanism claim. It is a representational transfer result over projection extremes with structure-matched controls, not a gold complaint dataset.

Larger Thesis

The useful loop is: real data exposes a messy failure mode; synthetic prompts isolate a clean abstraction; probes find a candidate internal signal; bridge tests expose transfer mismatch; real-data projection finds a narrower shape-specific signal; row reading improves the ontology.

TODO

Write the closing in your voice. Keep it practical: this is a template for using mech interp as an engineering feedback loop, not just as a benchmark result.

Next Steps

- Hand-label top and bottom rows with `current_action_size_conflict`, `retrospective_history_conflict`, `strategy_fulfillment_conflict`, `interpretation_or_rule_conflict`, and `unclear_or_label_mismatch`.
- Inspect neighboring settings cells after hand-labeling: L28, L32, and L36 `settings_end`.
- Find true non-complaint production controls.
- Defer causal tests and interventions until the label ontology is cleaner.